

Phaser Design Notes

Design Considerations

- User Physical Interaction
- Maintenance & Assembly
- Robustness & Reliability
- Appearance

Magazine Design Notes

- Make sure to include a USB port in the magazine so we can reprogram the Arduino
- One-way loading shape
- Make sure the magazine won't fall or bump out while in use
- Make it easy to insert and remove
- Have the body connection oriented downward with a 'lip' around the bottom part of the chamber.
 - The user places the magazine in the upper, wider part of the chamber, and slides it down into the narrower 'lipped' part to engage the connectors.
 - In this orientation, gravity will help the magnetic connectors keep the magazine in place
 - The lip helps to keep the magazine from falling out
 - Put a 'flange' on the top of the magazine so that the user can easily slide the magazine out

Trigger Notes

- Rather than a first-finger trigger, perhaps a thumb button on the rear shoulder of the phaser?
 - Easier for small hands to grip the handle without accidentally triggering
 - Less like a 'real gun'
 - Easier to assemble, more robust

Body Notes

- Make sure that the magazine chamber design keeps the magazine's charge panel highly visible
- Right-handed design - have the magazine chamber open to the left so that the charge panel is visible while the user is holding it
- Mirror-image design - Make sure that the magazine and the chamber are mirror-image symmetrical so we can print a left-handed body and use the same magazines
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